

Statistical Inference and Modeling

2102575

Suwichaya Suwanwimolkul¹

¹Electrical Engineering
Chulalongkorn University

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16. Clustering algorithms

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Supervised vs unsupervised learning

Supervised learning

$$\{X_j, y_j\}_{j=1}^m$$

Using $X_j \rightarrow \hat{y}_j$

Unsupervised learning

$$\{X_j\}_{j=1}^m$$

Using $X_j \rightarrow$ the data behavior

Examples of unsupervised learning

- ① data clustering
- ② dimensionality reduction:
 - data visualization
 - data compression
- ③ data association (rule-based machine learning)

Characteristics of unsupervised learning

- ① No ground truth $\rightarrow$ hard to assess the result from the model.
- ② More subjective $\rightarrow$ no simple goal for the analysis $\rightarrow$ usually based on rules.

Unsupervised learning algorithms

① Principle component analysis (PCA)

② k-mean clustering ←

③ hierachical clustering ←

④ GMM

⑤ DBSCAN

→ เราไม่สนใจเรื่อง Noise } Distance Base

↳ Robust ต่อ Noise เป็นพิเศษ } Density Base

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Target:

$$\begin{aligned}
 & \underset{C_1, C_2, \dots, C_K}{\operatorname{argmin}} \sum_{k=1}^K \frac{1}{|C_k|} \sum_{j, j' \in C_k} \|x_j - x_{j'}\|_2^2 = \sum_{k=1}^K \frac{1}{|C_k|} \sum_{j, j' \in C_k} \sum_{i=1}^n (x_{ji} - x_{j'i})^2 \\
 & \quad = \sum_{j \in C_k} \|x_j - \bar{x}_{C_k}\|_2^2 \\
 & \quad x_j = [x_{ji}]_{i=1}^n
 \end{aligned}$$

กำหนด Cluster

x ที่อยู่ใน Cluster เดียวกัน
 $\Rightarrow$ distance ใกล้เคียงกัน.

$\|x_j - x_{j'}\|_2^2$

K-mean clustering algorithms

Target:

$$\underset{C_1, C_2, \dots, C_K}{\operatorname{argmin}} \sum_{k=1}^K \frac{1}{|C_k|} \sum_{j \in C_k} \|\mathbf{x}_j - \bar{\mathbf{x}}_k\|_2^2 = \sum_{k=1}^K \frac{1}{|C_k|} \sum_{j \in C_k} \sum_{i=1}^n (x_{ji} - \bar{x}_{ki})^2$$

Algorithm

Initialization:

- Assume the number of clusters (K).
- randomly assign a number of 1 to K to each training sample.
- Randomly (or calculating) the centroid of cluster.

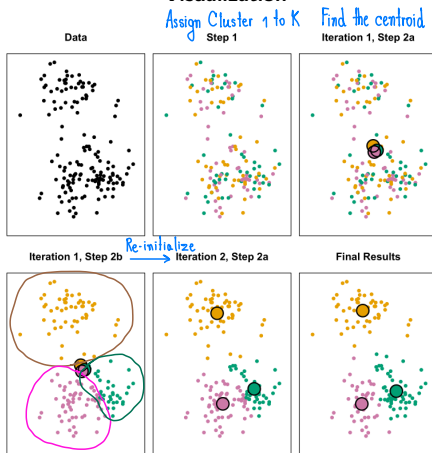
Start: iterate the following steps until the cluster assignments stop changing

- For each cluster, compute the cluster centroid $\bar{\mathbf{x}}_k$.

$$\bar{\mathbf{x}}_k = \frac{1}{|C_k|} \sum_{j' \in C_k} \mathbf{x}_{j'}$$

- Assign each training sample to the cluster whose centroid is the closest to the training sample.

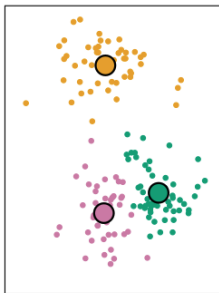
Visualization



K-mean clustering algorithms at inference

- Let $\bar{x}_k$ be the centroid obtained from the training phase.
- Given the input feature $x \in \mathcal{R}^n$, one can allocate the cluster of the input features by finding the cluster whose centroid is the closest to the input features.

$$\underset{k \in \{1, 2, \dots, K\}}{\operatorname{argmin}} \quad ||x - \bar{x}_k||_2^2$$



Example exercise: [https:](https://github.com/GenAI-CUEE/Statistical-Learning-EE575/blob/master/kmean_exercise.pdf)

[//github.com/GenAI-CUEE/Statistical-Learning-EE575/blob/master/kmean_exercise.pdf](https://github.com/GenAI-CUEE/Statistical-Learning-EE575/blob/master/kmean_exercise.pdf)

Hyper-parameters in the k-mean clustering algorithm

- The number of clusters (K).
- The distance function. l_1, l_2, l_∞ norm.
- How the cluster centroid is calculated.

Hierarchical clustering

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4 Hierarchical clustering

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Summary

Hyper-parameters in hierarchical clustering

Hierarchical clustering: making a dendrogram (agglomerative)

Algorithm

Initialization:

- Assign m training samples as m clusters.
- Measure the distances between each pair of clusters. For all $\binom{m}{2}$ pairs, there will be $\binom{m}{2}$ distance values.

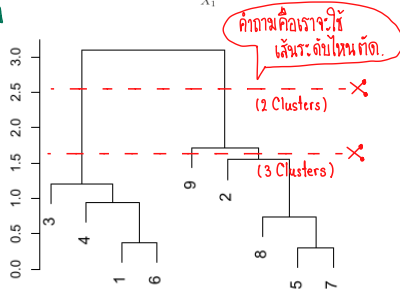
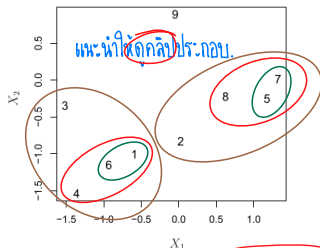
Start making the dendrogram:

For the number of clusters = $m, m-1, \dots, 2$

- Measure the distances of the rest number of clusters
- Fuse the similar pairs (this can be done using **single linkage**).
Point ที่ใกล้กันจะอยู่ด้วยกัน
- Record the similarity value as the height of the dendrogram
- Reduce the number of clusters by 1.

Assumption : ตอนแรกเลย เราจะมีว่า
1 sample = 1 cluster

Visualization

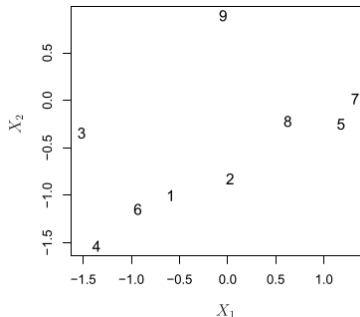


Making a dendrogram and distance matrix

At the number of cluster $K = m, m - 1, m - 2, \dots, 2$

- Measure the distances of K clusters (e.g., $K = m$)

	1	2	3	4	5	6	7	8	9
1	0.00000	0.989949	0.948683	1.656050	1.811077	1.019804	2.009975	1.204159	1.180042
2	0.989949	0.000000	1.649242	1.677051	1.208305	0.948683	1.581139	0.781025	1.650000
3	0.948683	1.649242	0.000000	1.154340	2.701851	0.989949	2.942788	2.109502	2.030394
4	1.656050	1.677051	1.154340	0.000000	2.884874	0.750000	3.250000	2.412986	2.830194
5	1.811077	1.208305	2.701851	2.884874	0.000000	2.154066	0.447214	0.608276	1.591383
6	1.019804	0.948683	0.989949	0.750000	2.154066	0.000000	2.505993	1.664332	2.147673
7	2.009975	1.581139	2.942788	3.250000	0.447214	2.505993	0.000000	0.854400	1.500833
8	1.204159	0.781025	2.109502	2.412986	0.608276	1.664332	0.854400	0.000000	1.162970
9	1.180042	1.650000	2.030394	2.830194	1.591383	2.147673	1.500833	1.162970	0.000000

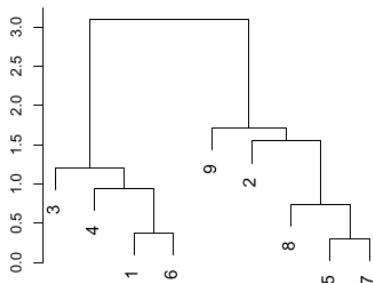


- Fuse the similar pairs (this can be done using single linkage).
- Record the similarity value as the height of the dendrogram.

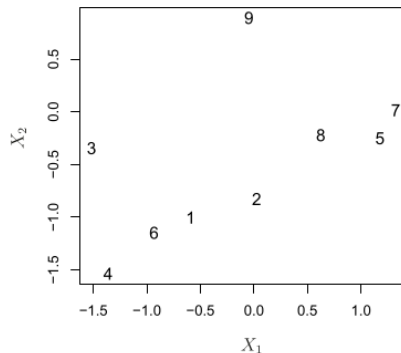
You can download and import the data from [Data](#). Then, see the hierarchical clustering using interactive illustration: [DataTab](#).

Hierarchical clustering with dendrogram

Dendrogram



Clustering



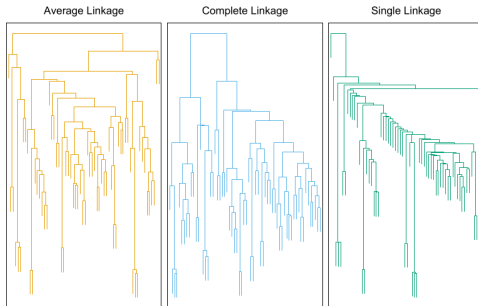
Hyper-parameters

- Single Linkage
- Euclidean distance

Linkage types

Linkage	Description
Complete	Maximal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the <i>largest</i> of these dissimilarities.
Single	Minimal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the <i>smallest</i> of these dissimilarities. Single linkage can result in extended, trailing clusters in which single observations are fused one-at-a-time.
Average	Mean intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the <i>average</i> of these dissimilarities.
Centroid	Dissimilarity between the centroid for cluster A (a mean vector of length p) and the centroid for cluster B. Centroid linkage can result in undesirable <i>inversions</i> .

TABLE 12.3. A summary of the four most commonly-used types of linkage in hierarchical clustering.



Hierarchical clustering idea

What is it?

Cluster via Dendrogram

- Hierarchical clustering **relies on the nested relationship between data samples**. One can divide the data into multiple level of (nested) similarity and dissociation.
- A **tree-based** representation, called dendrogram, is used to represent the nested relationship, which can be interpreted as the data clustering.
- **Note: the dendrogram is different from the tree-based classifiers in the previous chapter.**

Good vs bad

- (+) Use the nested relationship to **divide the data** into multiple levels of clustering.
- (+) Don't need to assume the number of clusters (K). The maximum number of clusters is equal to the number of training samples (m).
- (-) If the data features do not exhibit the nested relationship, the hierarchical clustering will perform poorly.
- (-) Can be **sensitive to outliers**. To fix this, you can try to exclude the outliers or choose to use the soft-method for hierarchical clustering.

Hyper-parameters

- What type of linkage should be used?
- What dissimilarity measure should be used?
- Where should we cut the dendrogram in order to obtain clusters?
- Should the observations or features first be standardized in some way?

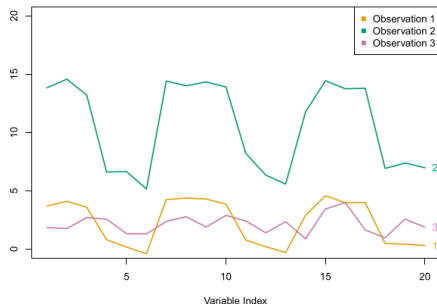


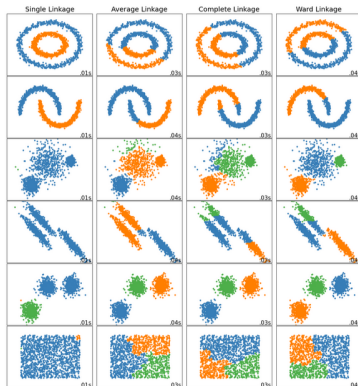
FIGURE 12.15. Three observations with measurements on 20 variables are shown. Observations 1 and 3 have similar values for each variable and so there is a small Euclidean distance between them. But they are very weakly correlated, so they have a large correlation-based distance. On the other hand, observations 1 and 2 have quite different values for each variable, and so there is a large Euclidean distance between them. But they are highly correlated, so there is a small correlation-based distance between them.

Hierarchical clustering algorithms at inference

- Let $\bar{x}_k$ be a representative features of the clusters from the training phase.
- Given the input feature $x \in \mathcal{R}^n$, one can allocate the cluster of the input features by finding the cluster whose $\bar{x}_k$ is the closest to the input features.

$$\underset{k \in \{1, 2, \dots, K\}}{\operatorname{argmin}} \quad ||x - \bar{x}_k||_2^2$$

เราต้อง Assume Cluster
↓
เพราะเราไม่รู้มันพอที่เรา.



Agglomerative clustering

- [1] Gareth James et al. *An Introduction to Statistical Learning: with Applications in R*. Springer, 2013. URL: https://hastie.su.domains/ISLP/ISLP_website.pdf.download.html.